

JEFFRIN SINGH R

AI ENGINEER | COMPUTER VISION | BACKEND SYSTEMS

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[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

AI Engineer focused on building real-time computer vision pipelines and scalable backend systems. Experienced in developing production-ready applications using YOLO, InsightFace, OCR, and ONNX Runtime, with proven improvements in latency (~30%) and real-time processing (~20 FPS).

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

AI / Computer Vision: YOLO, InsightFace, OpenCV, OCR (PaddleOCR/EasyOCR), ONNX Runtime, ByteTrack

Backend: FastAPI, Flask, REST APIs, WebSockets

Frontend: React, HTML, CSS, Tailwind CSS

Databases: PostgreSQL, SQL Server, MySQL

Tools & DevOps: Docker, Git, Vercel, Render

EXPERIENCE

AI Engineer Trainee

Feb 2026 – April 2026

[Invendoi AI Solutions](#), Kochi

- Built a real-time ANPR system using YOLO + OCR + ByteTrack for vehicle detection and tracking in live video streams
- Reduced inference latency by ~30% using ONNX Runtime optimization
- Achieved ~20 FPS on 720p streams, enabling real-time deployment
- Improved tracking accuracy and minimized duplicate detections using ByteTrack
- Designed scalable pipelines for multi-camera surveillance systems deployed in real-world scenarios
- Developed and deployed production-grade platforms used by real users: [Invendoi](#) • [Invendo AI Academy](#) • [Campus Platform](#)
- Built and optimized frontend using React + Vite + Tailwind CSS for performance and responsiveness
- Integrated Gemini API chatbot to handle real-time user queries and support

KEY PROJECTS

AI Face Recognition System

- Built real-time face recognition pipeline using InsightFace + cosine similarity
- Achieved <100ms response latency
- Developed FastAPI backend with PostgreSQL logging system
- Designed for multi-camera deployment using Docker

Automatic Number Plate Recognition (ANPR) System

- Developed end-to-end pipeline using YOLO + ByteTrack + OCR
- Improved OCR accuracy using preprocessing under real-world conditions
- Designed system for real-time surveillance applications

Secure Payment & Subscription System

- Built Flask-based web app with Razorpay integration
- Implemented authentication, subscription validation, and payment workflow
- Designed database schema for user and payment management

EDUCATION

B.E. – Computer Science and Engineering

2026

Loyola Institute of Technology and Science

CERTIFICATIONS

- AI/ML Training Program – Feather Softwares
- Web Development Internship – Roriri Software Solutions
- Python for Data Science – Udemy
- Bootstrap Mastery – Udemy